CAMPUS MAKERSPACE MANAGEMENT SYSTEM

A platform for managing shared equipment: 3D printers, laser cutters, electronics workstations

- *Users view equipment, reserve time slots and track availability*
- *Staff manage access, maintenance and consumable stock*

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Agenda: Stakeholders → User stories → Use cases → Class & sequence diagrams → Requirements analysis

STAKEHOLDERS

Category	Stakeholders
Primary (direct users)	Students, Staff
Secondary (use outputs indirectly)	-
External (regulators)	Data protection regulation (GDPR)
Hidden (maintainers)	IT

STAKEHOLDER ROLES

Roles

- Students: requirement provider, validator
- Staff: requirement provider, constraint source, validator, decision-maker, domain expert
- IT: maintainer / operator, requirement provider
- GDPR: constraint source

Conflicts

- Students want long, flexible bookings ↔ Staff need fair use and maintenance time → Staff rules: booking limits, check-in
- Students want quick access ↔ IT needs security → single university login

Three user stories drive the core flows

Reserve a 3D printer

Reserve a printer for a time slot without waiting in line.

• *Tasks*: view availability calendar · book a slot · check in · get confirmation

• *MVP*: availability calendar · booking without overlapping · check-in within 15 minutes, otherwise the slot is released

• *Release* : email confirmation · maximum booking duration

Report an equipment issue

Report a malfunction or damage straight from the platform.

• *Tasks*: select a reserved machine · describe the problem · add a photo · get confirmation

• *MVP*: report only for machines the user has reserved · text description · staff see the report

• *Release* : photo upload · email confirmation · urgency levels · moderation against abuse

Manage equipment maintenance

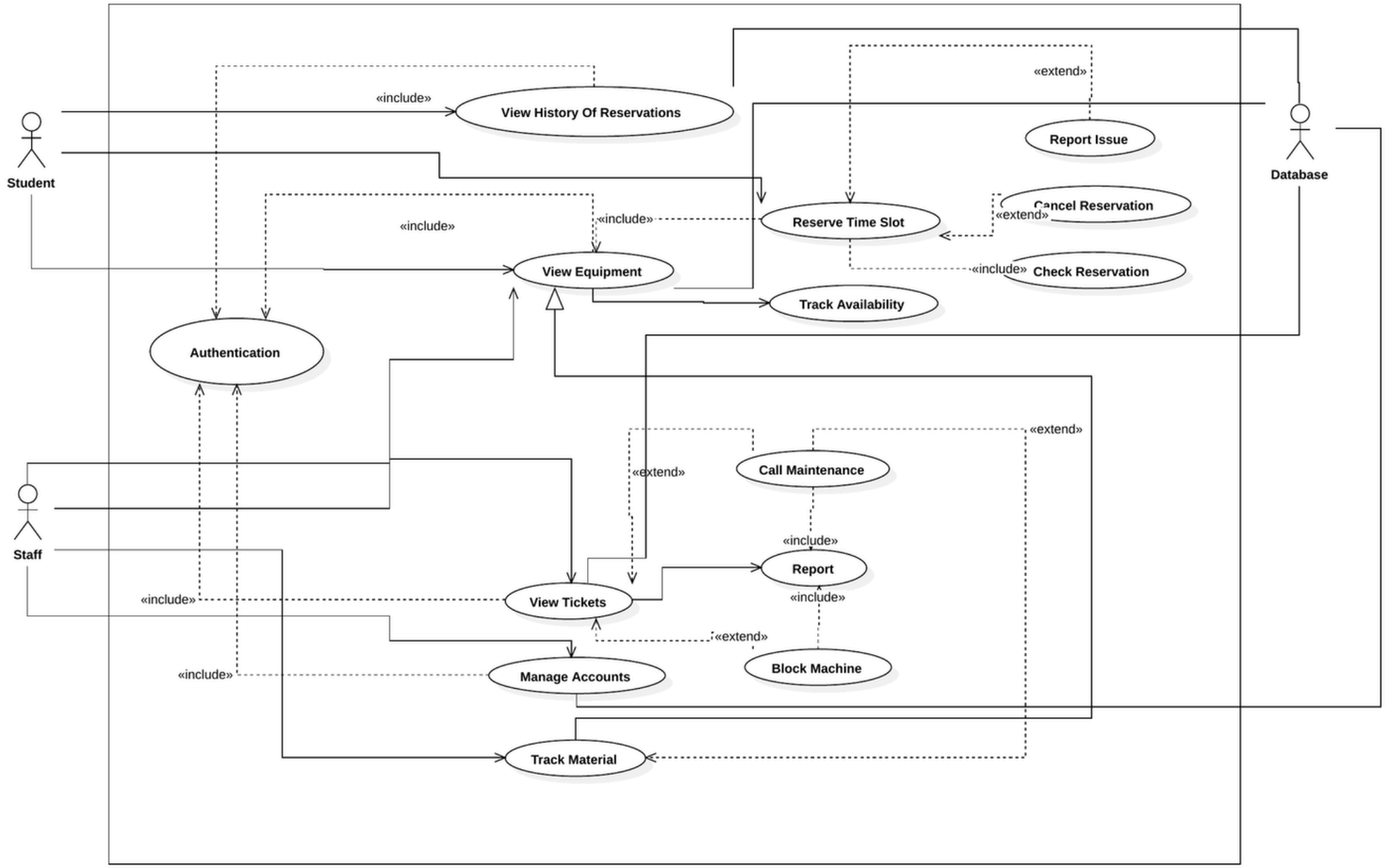
Mark a machine as under maintenance so nobody books it while broken.

• *Tasks:* open a report · change its status · block the machine · notify affected users

• *MVP:* change report status · block the machine · cancel its bookings and notify users

• *Release* : move bookings to another machine · unblock a machine · email templates

USE CASE DIAGRAM



Actors

- Student: books machines and reports issues
- Staff: handles tickets, machines, stock and accounts
- Database (secondary): stores accounts, bookings and tickets

14 use cases

- Booking: view equipment, track availability, reserve a slot, check or cancel it, report an issue, view history
- Maintenance: view tickets, report, block machine, call maintenance, track material
- Access: authentication, manage accounts

INCLUDE & EXTEND

How the 14 use cases are connected, and why

«include»

always runs as part of the base use case

- View Equipment, View History, View Tickets and Manage Accounts include Authentication: login rules live in one place
- Reserve Time Slot includes View Equipment and Check Reservation: pick a machine first, confirm 15 min after the start
- Block Machine and Call Maintenance include Report: every staff action is documented

«extend»

optional: runs only in a specific situation

- Cancel Reservation and Report Issue extend Reserve Time Slot: only possible once a booking exists
- Block Machine and Call Maintenance extend View Tickets: an optional staff reaction to a ticket
- Call Maintenance extends Track Material: refill when stock drops below the minimum

Generalization: Track Material is a special View Equipment (the equipment view plus stock data)

USE CASE DESCRIPTIONS

One template for all 14 use cases

1. Name
2. Actors (primary / secondary)
3. Preconditions
4. Main success scenario
5. Extensions (alternate flows)
6. Postconditions

Example: Check Reservation

1. System reminds the user 15 min before the start
2. User opens the reservation
3. System checks that the status is "Awaiting confirmation"
4. System shows equipment, time slot and status
5. User confirms the reservation
6. System sets the status to "Confirmed"
7. User receives a confirmation

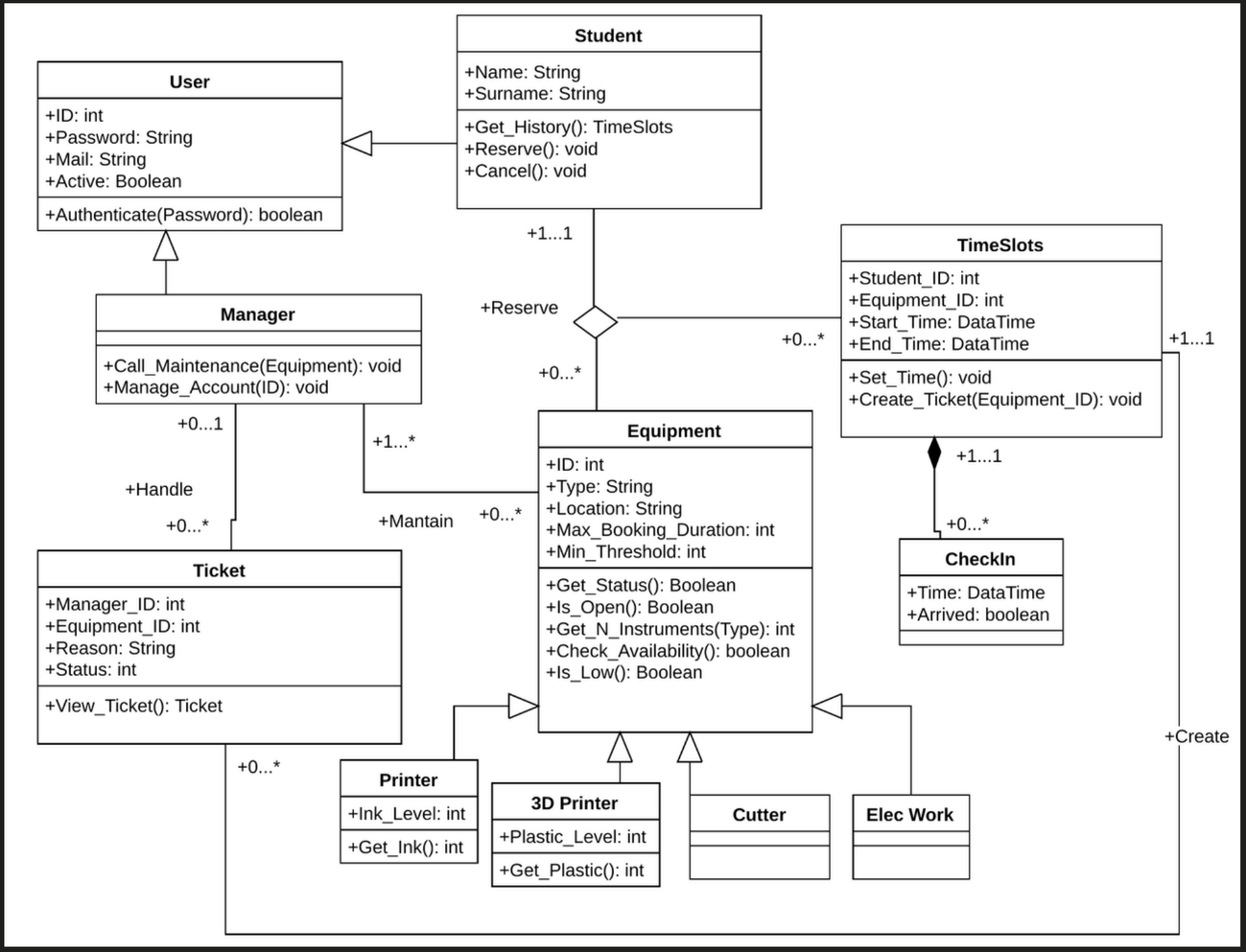
Extensions, numbered by step

3a. Cancelled by staff: the user sees the reason and cannot confirm
5a. Not confirmed before the start: the booking is cancelled, the slot is released and the user is notified

Postconditions

Success: the reservation is confirmed
Exception: the reservation is cancelled and the user is notified

CLASS DIAGRAM

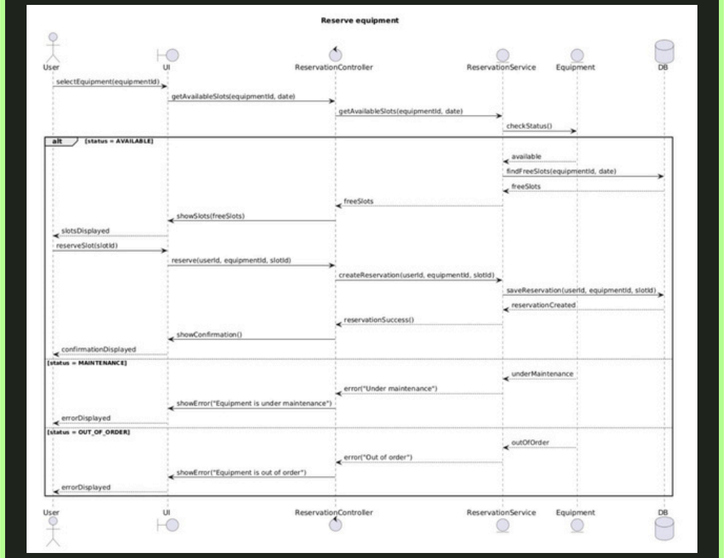


Key design decisions

- User is the superclass of Student and Manager: one login, different rights.
- TimeSlots is the reservation: who, which machine and when.
- CheckIn is a composition, no reservation, no check-in. Arrived supports the no-show rule.
- A Ticket is created from a TimeSlots: only users who booked a machine can report it. (A manager is attached later)
- Machine types are subclasses of Equipment: a new type does not change the booking logic, and this remedy allow us to upscale it in the future.

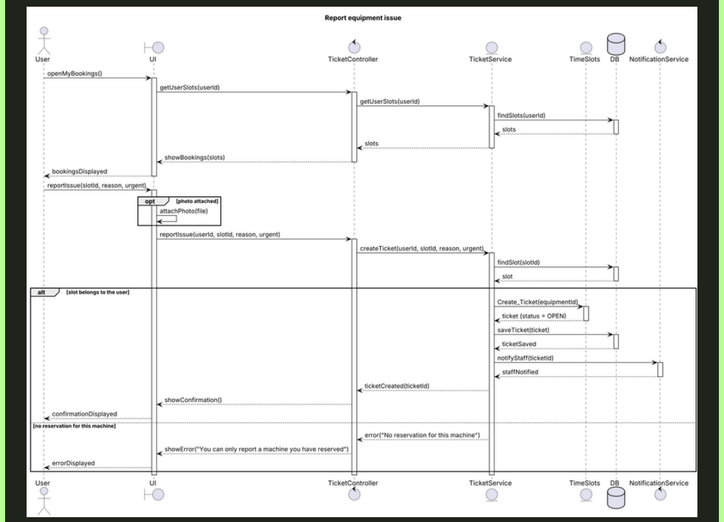
SEQUENCE DIAGRAMS

Three core interactions, one for each user story



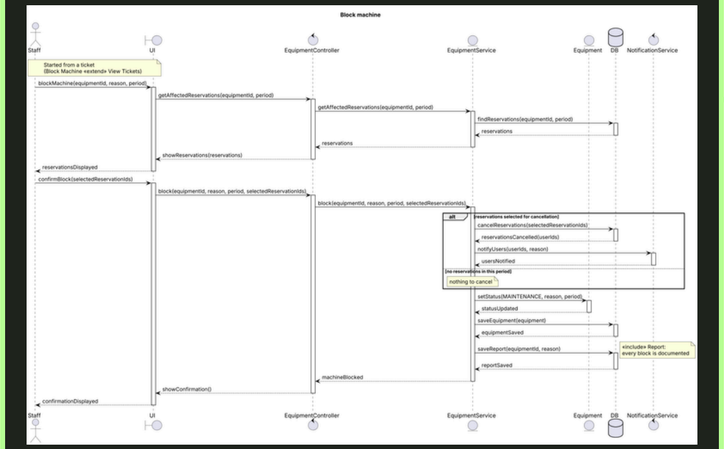
1. Reserve time slot

Shows free slots and saves the booking.
alt: machine under maintenance or out of order



2. Report issue

Creates a ticket from the user's time slot and notifies staff.
alt: no reservation for this machine



3. Block machine

Cancels selected bookings, sets MAINTENANCE and saves a report.
alt: no bookings in the period

REQUIREMENTS ANALYSIS

FUNCTIONAL REQUIREMENTS

What the system does

- Reservation: a user books a free time slot; the same slot can never be booked twice.
- No-shows: a booking must be confirmed at most 15 minutes after the start, otherwise the slot is released.
- Issue reporting: a user can only report a machine they have reserved.
- Maintenance: staff block a faulty machine, and affected users are notified automatically.
- Stock: staff are warned when a consumable drops below its minimum level, like 3D printer filament, is running low, so they can reorder it in time.

NON-FUNCTIONAL REQUIREMENTS

How well the system works

- Scalability: supports 1,000 students.
- Performance: the availability calendar loads in under 2 seconds.
- Security: only the Staff role can block machines and manage accounts; login is locked after 3 failed attempts.
- Privacy (GDPR): students see only their own data; passwords are never stored in plain text.
- Reliability: no double booking, even when two users book the same slot at the same moment.

THANK YOU

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